

VALLEY HEAD 1 SSW, AL, USA

WMO: 720011

Lat: 34.5653N

Lon: 85.6171W

Elev: 311

StdP: 97.65

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.3	(c) -7.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 11.0	(c) 32.9	(d) N/A	(e) 31.8	(f) N/A	(g) 30.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-12.6	34.3	2.6	1.8	-14.4	35.6	-15.9	36.7	-17.4	37.7	-19.3	39.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	3.8	5.5	10.5	14.9	19.4	23.4	24.6	24.7	21.7	15.3	9.1	6.7
	DBStd	8.54	6.01	5.36	5.11	4.22	3.57	2.04	1.89	2.16	3.11	4.69	4.69	5.39
	HDD10.0	633	208	144	58	9	0	0	0	0	9	71	134	
	HDD18.3	1968	450	359	246	116	33	1	0	0	9	116	277	361
	CDD10.0	2466	16	18	72	156	291	402	452	455	352	174	45	31
	CDD18.3	759	0	0	2	13	66	153	193	197	111	23	1	1
	CDH23.3	6148	0	1	32	144	555	1297	1500	1608	839	167	5	0
	CDH26.7	2106	0	0	2	10	123	476	547	640	277	31	1	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	1440	124	114	164	127	121	116	141	92	118	81	127	112
	PrecMax	1772	244	345	403	277	286	254	324	175	280	252	262	266
	PrecMin	1058	33	14	48	30	27	14	24	15	20	2	33	21
	PrecStd	210	52	58	85	58	61	60	70	44	76	56	54	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	21.7	26.5	27.9	30.7	33.2	33.6	35.2	33.0	29.5	23.6	21.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	17.6	19.1	23.9	26.5	29.4	32.0	32.2	33.3	31.4	27.0	20.9	19.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	15.7	17.3	21.7	25.1	28.1	30.9	31.0	31.7	30.0	25.2	19.4	17.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	13.6	15.2	19.4	23.5	26.8	29.7	29.9	30.3	28.1	23.3	17.7	15.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.0	12.6	13.4	12.1	11.8	10.5	11.0	11.4	12.6	13.1	10.3
		MCDBR	11.5	14.3	14.9	15.0	14.0	13.7	12.0	12.9	13.7	13.6	14.2	10.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.314	0.319	0.356	0.393	0.445	0.481	0.533	0.530	0.424	0.372	0.337	0.314
	taud		2.508	2.490	2.412	2.295	2.191	2.124	1.995	1.991	2.280	2.413	2.458	2.528
	Ebn at Noon		896	930	917	893	846	812	767	762	835	856	858	875
	Edh at Noon		83	94	111	131	147	157	177	175	125	101	87	77
(44) All-Sky Solar Radiation	RadAvg		2.48	3.01	4.13	5.37	5.90	6.20	5.88	5.48	4.72	3.88	2.82	2.11
	RadStd		0.18	0.39	0.38	0.35	0.46	0.47	0.38	0.29	0.44	0.49	0.32	0.21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+145	+105

Nomenclature: See separate page